

Multicast over Swisscom SRv6 backbone

22 Sept. 2026
Mihai Ciurea
Swisscom, Network Connectivity,
Architecture&Engineering

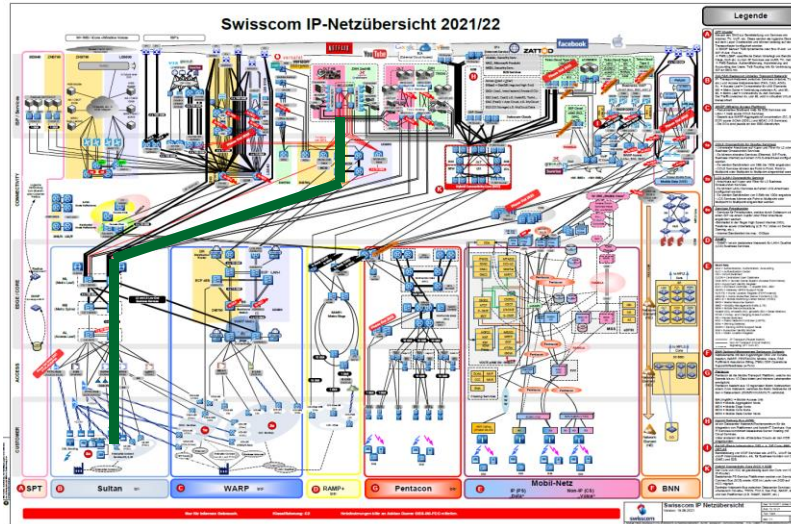
C1 Public



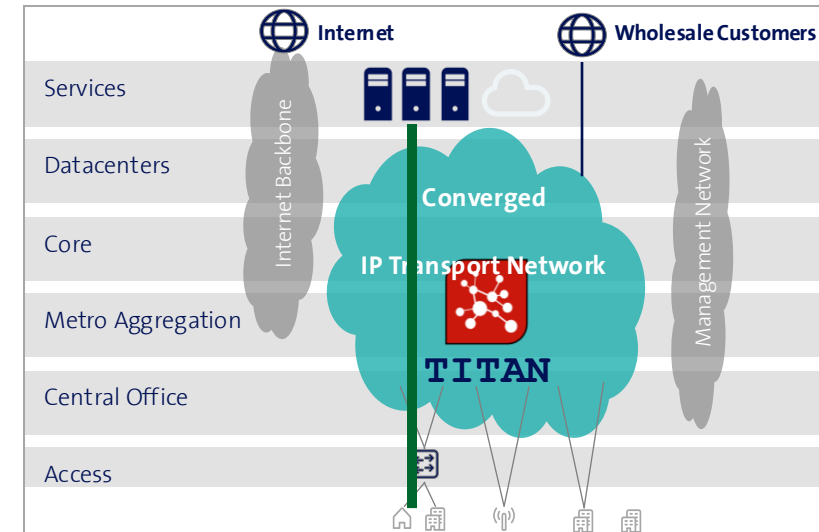


Swisscom's converged IP Network Vision ...continued

Swisscom IP Transport Network Landscape 2021



Network Vision 2026



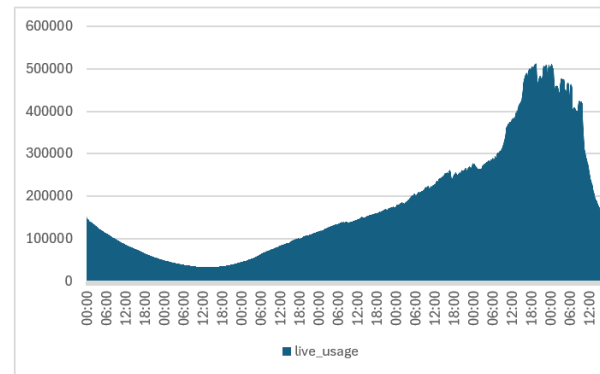
IP-TV Multicast



Today facts & figures

Swisscom offers today IP multicast for the following products:

- IP-TV
 - Serving 1.5M TV clients (Blue TV)
 - IP-TV wholesale (Other ISPs)
- Multicast for Business L3VPN



Peak concurrent live TV users (big sports event): ~0.5M

-> Serving these customers with unicast instead of multicast would increase the traffic by a factor of 5.

Mature, stable and successful products

Video Bitrates:
SD ≈ 3.5Mbps
HD ≈ 8.0Mbps
UHD ≈ 20Mbps

“Announced and long awaited, Swisscom’s offering of television down the telephone line was launched in November 2006 as its Bluewin TV service. In particular, Swisscom’s customers can access over 100 television channels and 70 radio stations.”

Source: ComCom’s Annual Report 2006





Requirements



No changes Source and Receiver side

- **2000** IPv4 multicast groups
- Group membership: IGMPv2
- Multicast protocol: PIM-SM
- Rendez-vous point: Anycast RP
- Source discovery: MSDP

Wish list:

- Use TITAN core to transport IP-TV
- **Do not introduce IPv4 in the new IPv6/SRv6 network**
- Proven, stable technology
- Simplicity and economics



- IPv6 PIM-SM/SSM is

- ... bridging the gap ...
enabling IPv4 multicast encapsulation in
IPv6 multicast

- Use TITAN core to transport IP-TV ✓
- **Do not introduce IPv4 in the new IPv6/SRv6 network** ✓
- Proven, stable technology ✓
- Simplicity and economics ✓✓

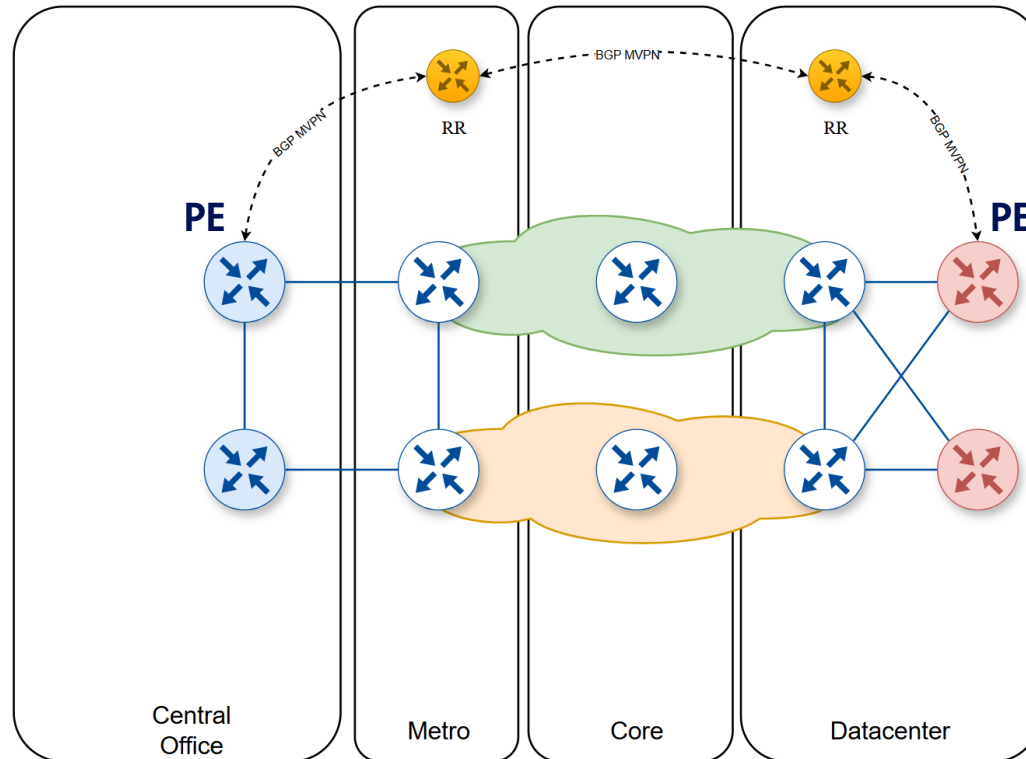




Customer/Provider Multicast

I-PMSI: inclusive multicast service interface

S-PMSI: selective multicast service interface



Tunnel type:

Value	Meaning
0x00	no tunnel information present
0x01	RSVP-TE P2MP LSP
0x02	mLDP P2MP LSP
0x03	PIM-SSM Tree
0x04	PIM-SM Tree
0x05	BIDIR-PIM Tree
0x06	Ingress Replication
0x07	mLDP MP2MPLSP
0x08 - 0xFA	Unassigned
0xFB - 0xFE	Experimental
0xFF	Reserved

Route Types for A-D routes:

- + 1 - Intra-AS I-PMSI A-D route;
- + 2 - Inter-AS I-PMSI A-D route;
- + 3 - S-PMSI A-D route;
- + 4 - Leaf A-D route;
- + 5 - Source Active A-D route.

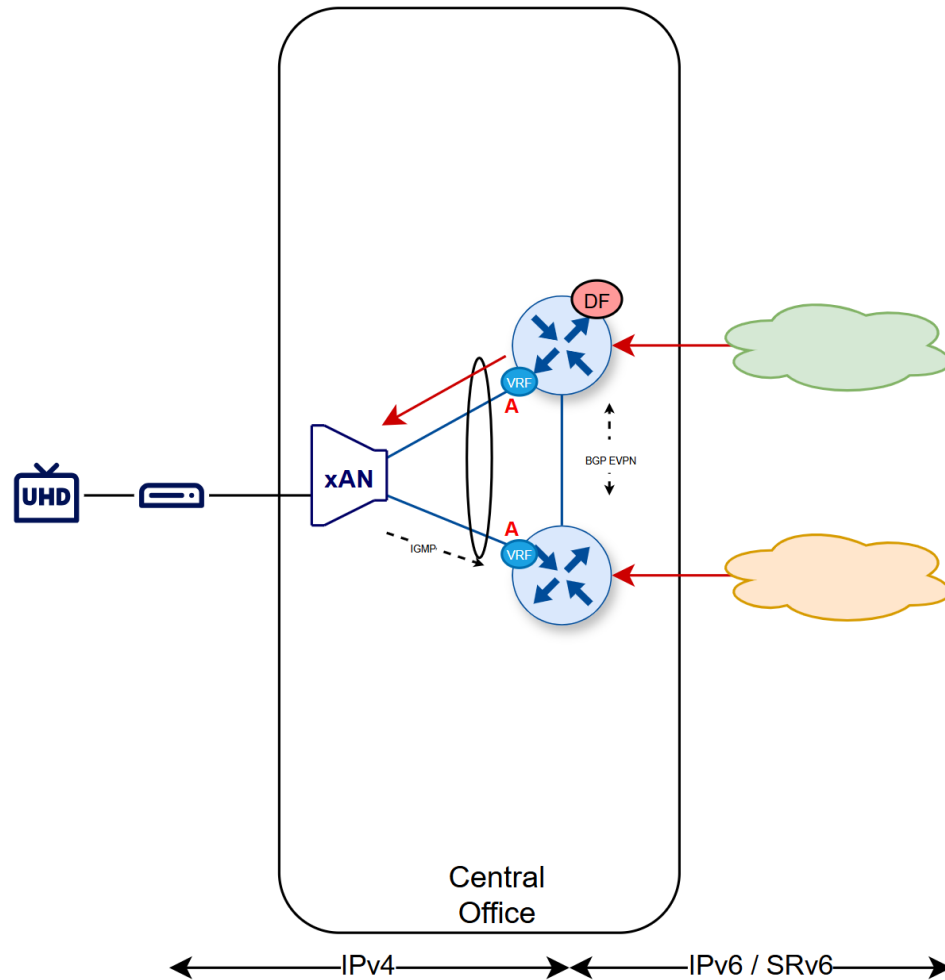
Route Types for C-multicast routes:

- + 6 - Shared Tree Join route;
- + 7 - Source Tree Join route;

```
multicast-routing
vrf TITAN_INT_TEST-00958-00
address-family ipv4
 mdt source ipv6 Loopback0
 rate-per-route
 interface all enable
 accounting per-prefix
 bgp auto-discovery pim6
 !
 mdt default ipv6 ff33::1:1
 mdt data ipv6 ff33:100::/116
```



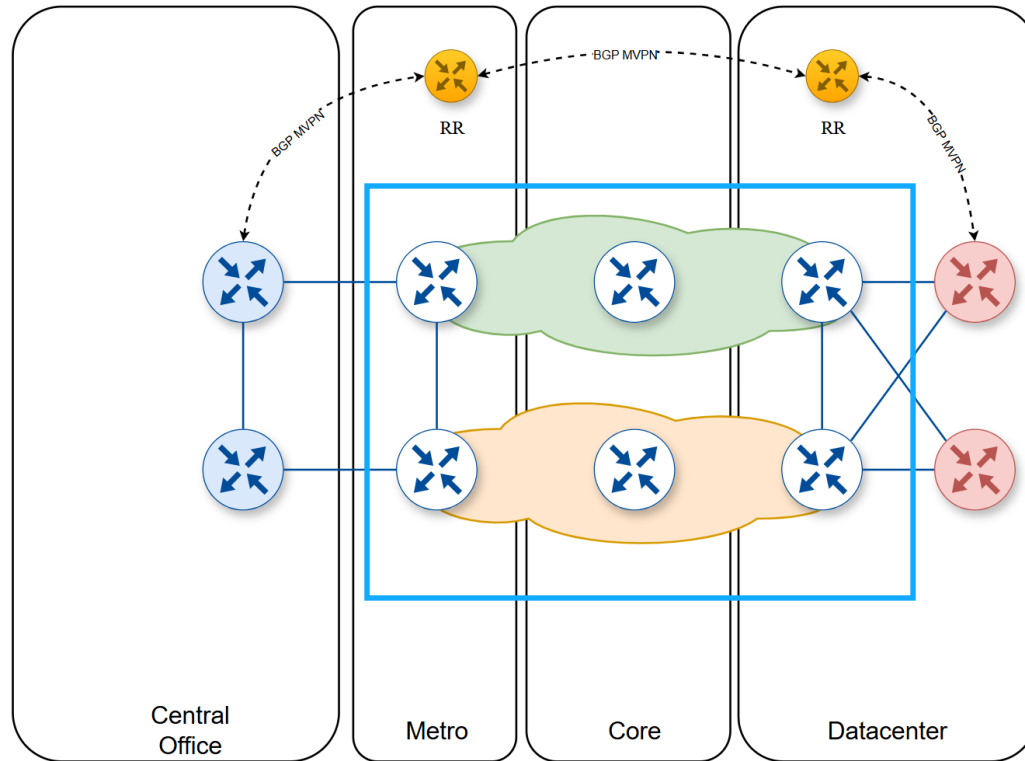
Access



- Multi-Homed Device A/A
- Bundle sub-interface in VRF - IPv4 address
- EVPN route-sync
- DF election is done based on bgp router-id and EVI
- BGP IPv4 mvpn address-family
- Core Tree pim-default-v6



Core



P Multicast	
src = fd7c:3f00:40f::1	source PE Loopback
dst = ff3e::1:1	default or data MDT
C Multicast	
src = 33.2.2.10	
dst = 239.186.77.77	
Payload	

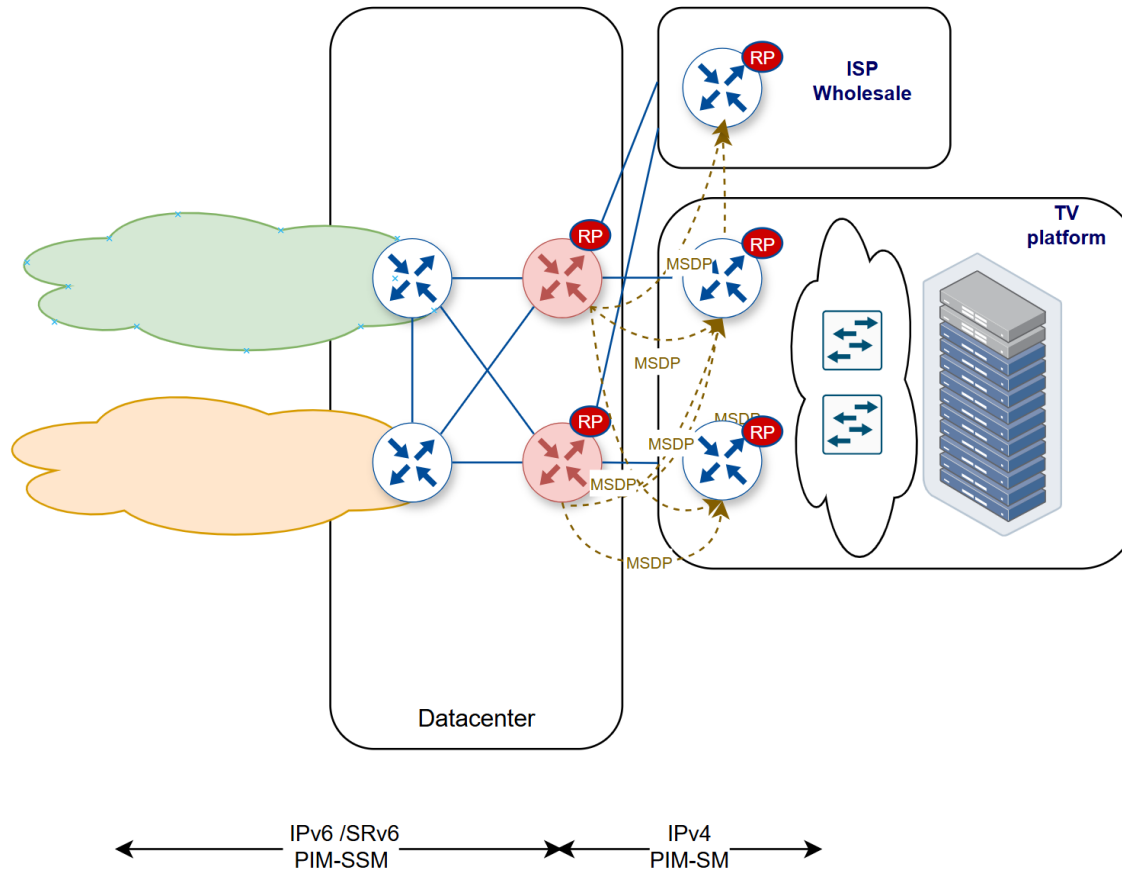
- Only multicast-routing enabled!
- IPv6 addressing for PIM-SSM:

```
| 8 | 4 | 4 | 8 | 8 | 64 | 32 |
+---+---+---+---+---+---+
|11111111|flgs|scop|reserved| plen | network prefix | group ID |
+---+---+---+---+---+---+
                        0      0      0
```

flgs	scop	NAME	REFERENCE
0	0	Reserved	[RFC4291], RFC 7346
1	1	Interface-Local scope	[RFC4291], RFC 7346
2	2	Link-Local scope	[RFC4291], RFC 7346
3	3	Realm-Local scope	[RFC4291], RFC 7346
4	4	Admin-Local scope	[RFC4291], RFC 7346
5	5	Site-Local scope	[RFC4291], RFC 7346
6	6	Unassigned	
7	7	Unassigned	
8	8	Organization-Local scope	[RFC4291], RFC 7346
9	9	Unassigned	
A	A	Unassigned	
B	B	Unassigned	
C	C	Unassigned	
D	D	Unassigned	
E	E	Global scope	[RFC4291], RFC 7346
F	F	Reserved	[RFC4291], RFC 7346



Headend



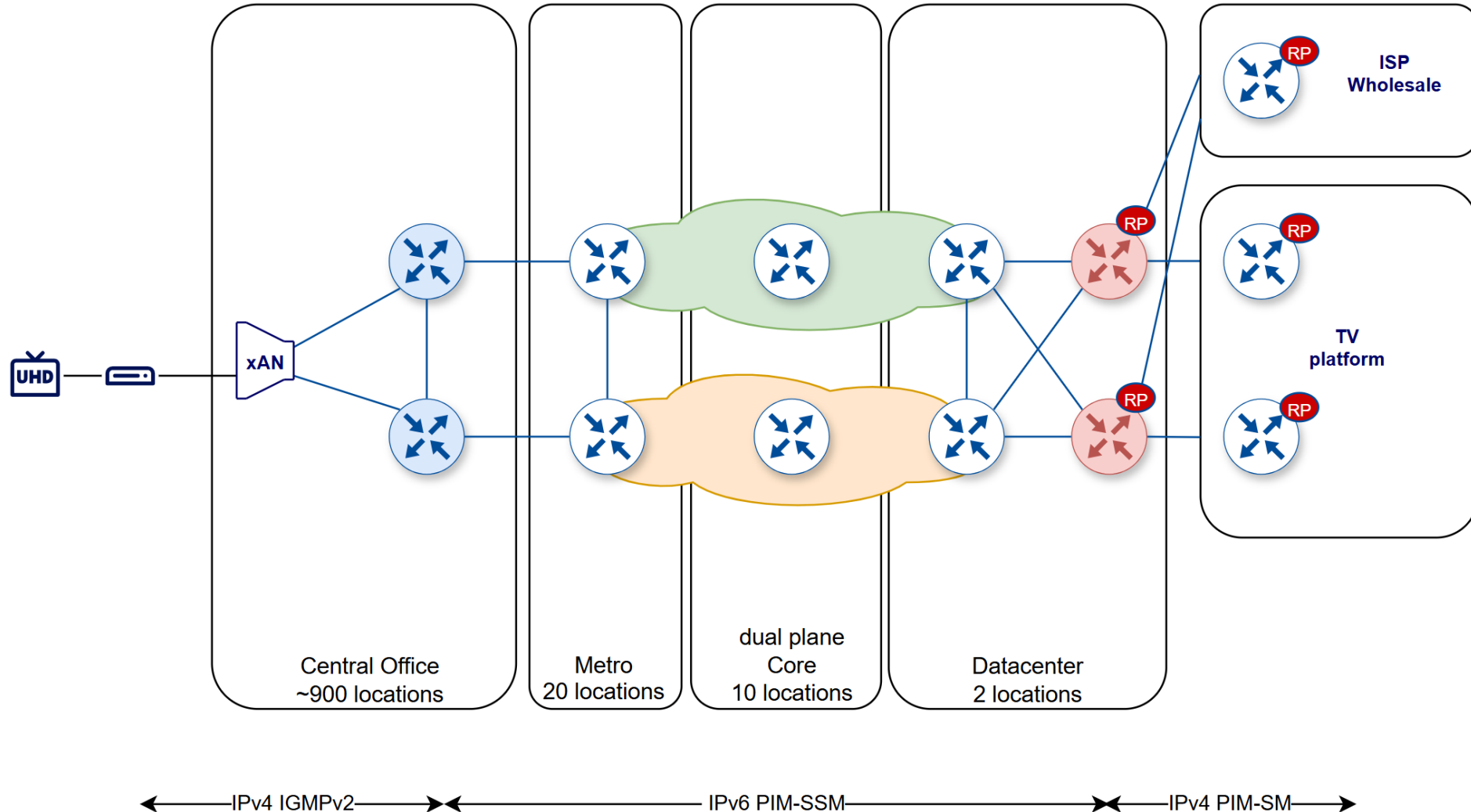
- The TV platforms advertise the sources to datacenter PE via MSDP
- BGP IPv4 mvpn address-family
- «default MDT»: BGP MVPN Route-Type 1
 - Receiver side PEs are configured as receiver-site (empty SRC and empty default MDT)
- «data MDT»: BGP MVPN Route-Type 3

*> [3][32][33.2.2.10][32][239.186.77.77][fd7c:3f00:40f::1]/216



Multicast flow setup

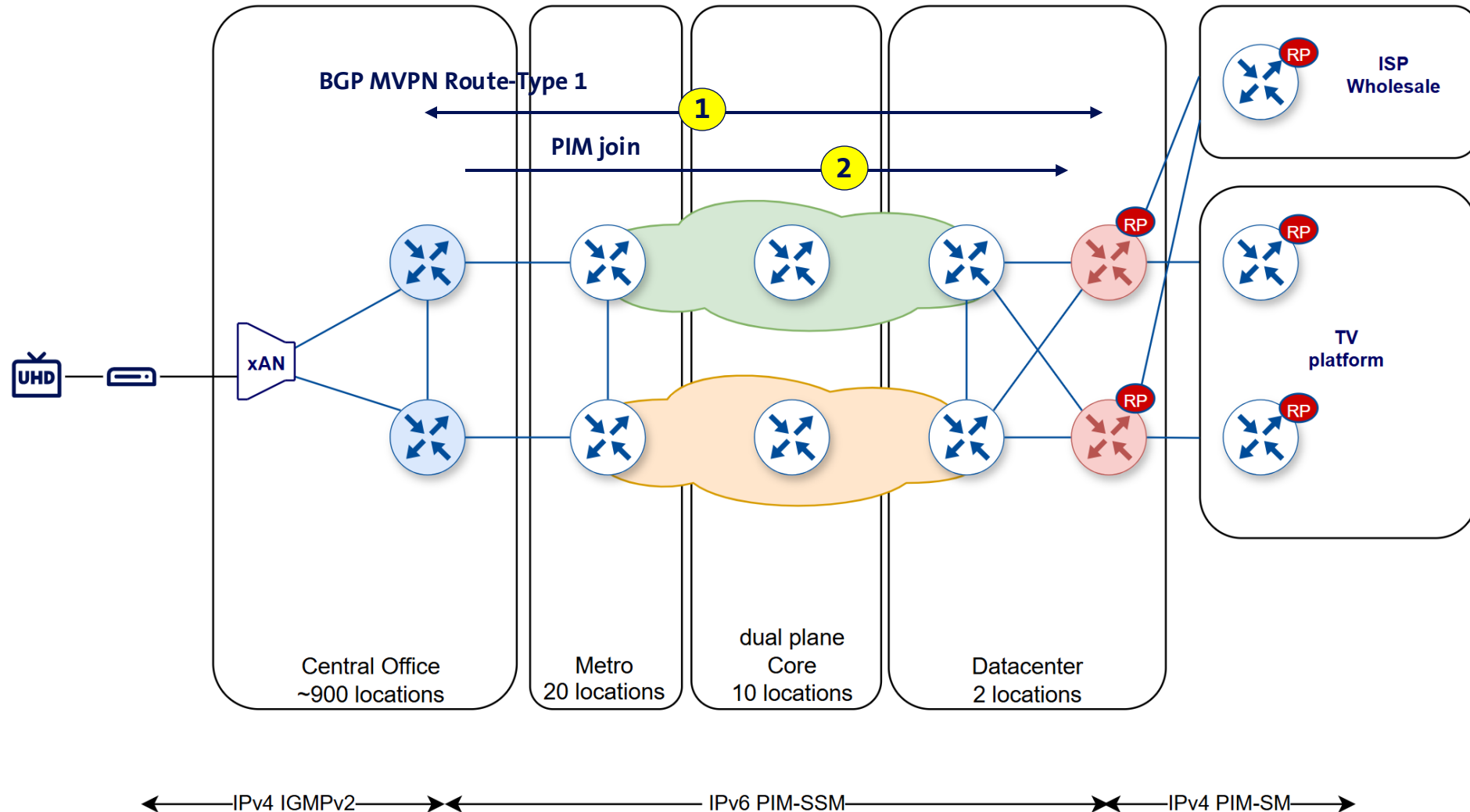
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Multicast flow setup – 1-2

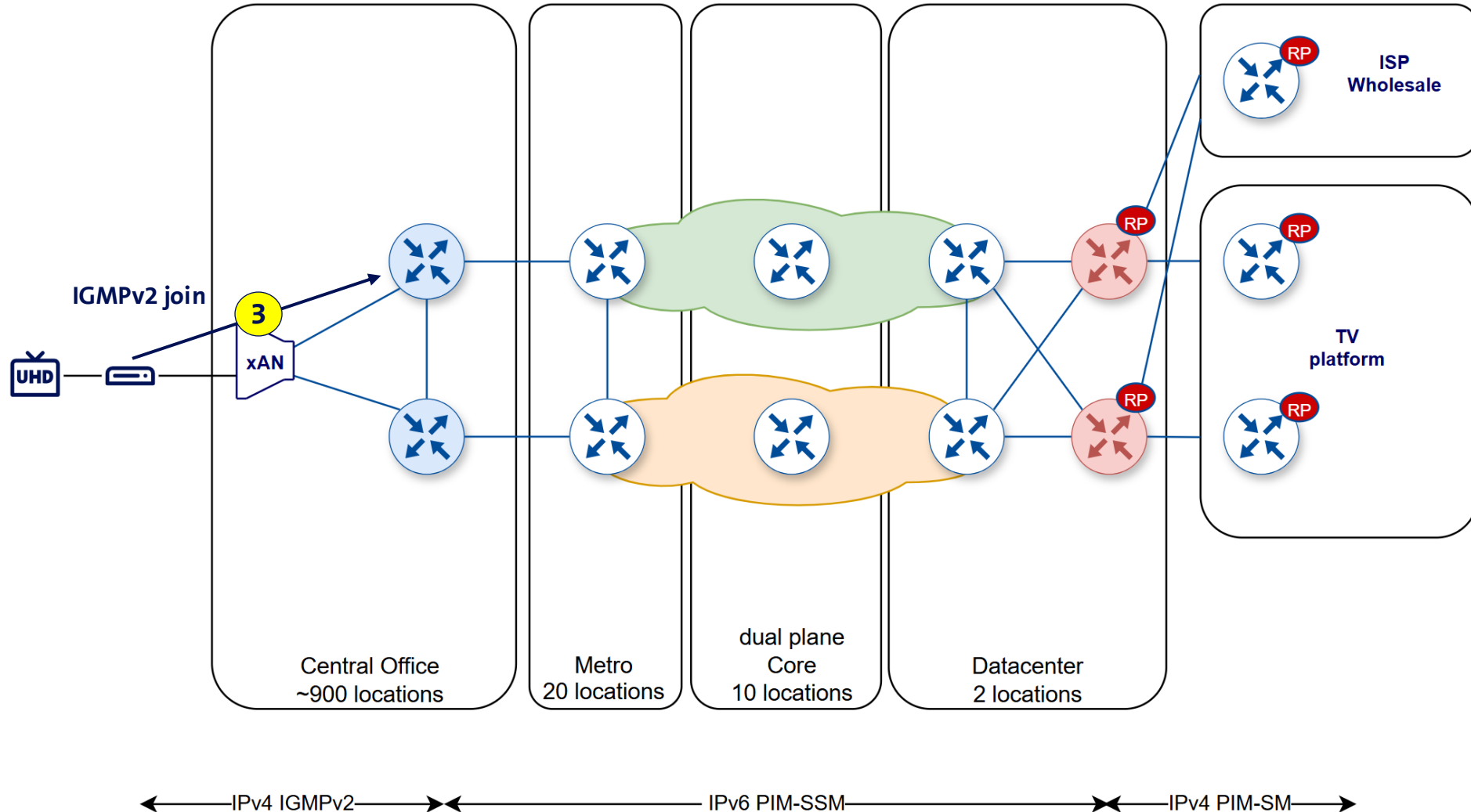
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Multicast flow setup - 3

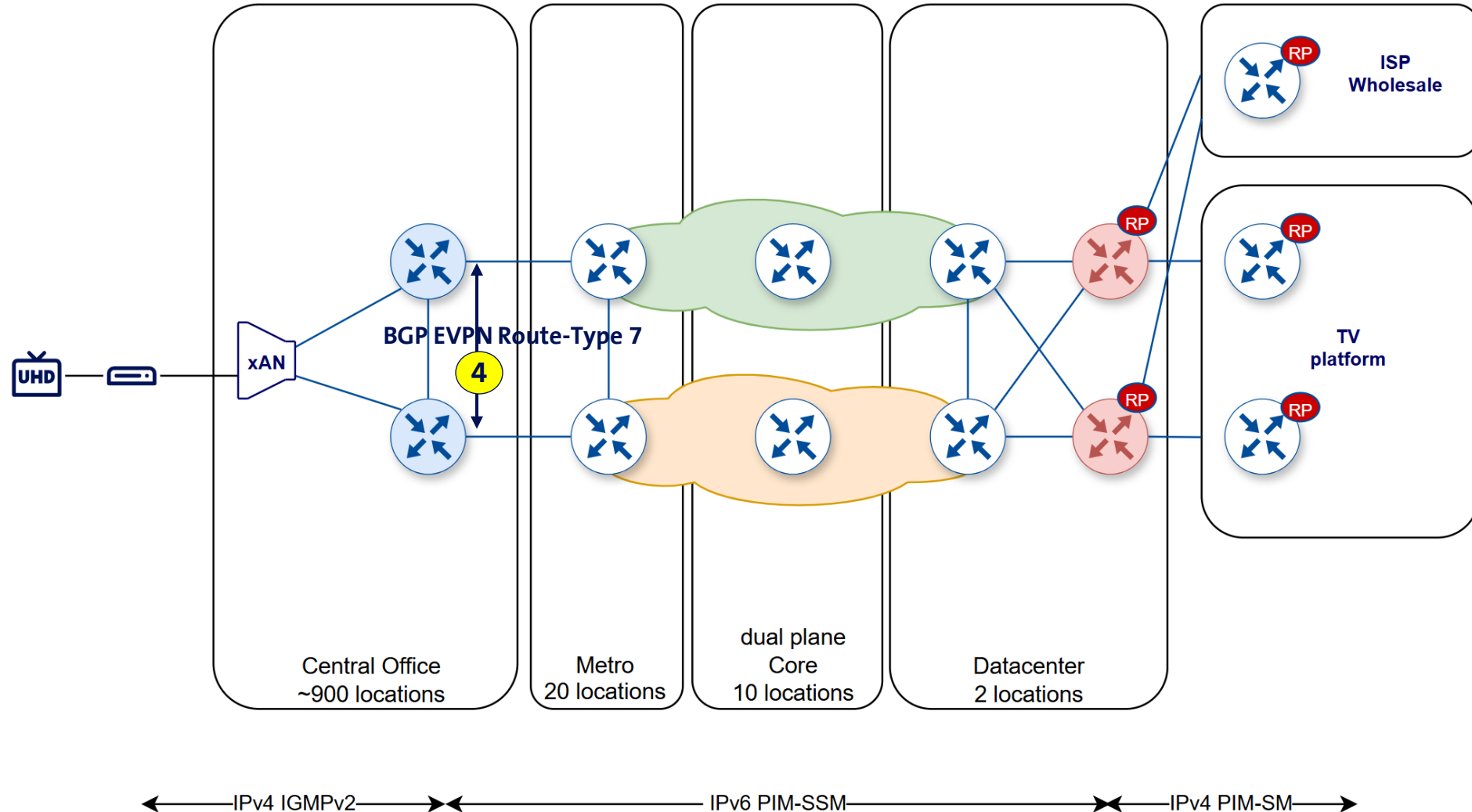
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Multicast flow setup - 4

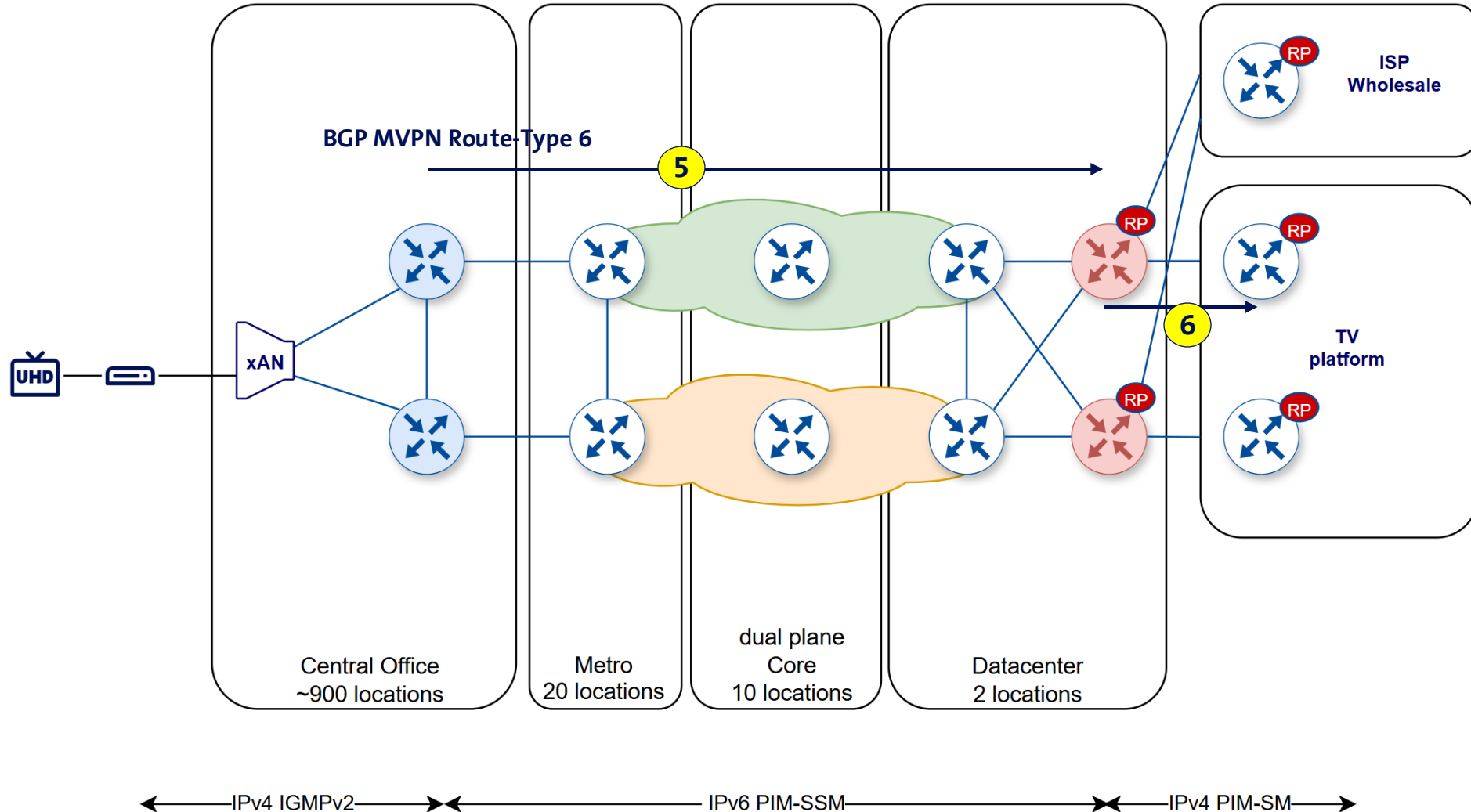
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Multicast flow setup – 5-6

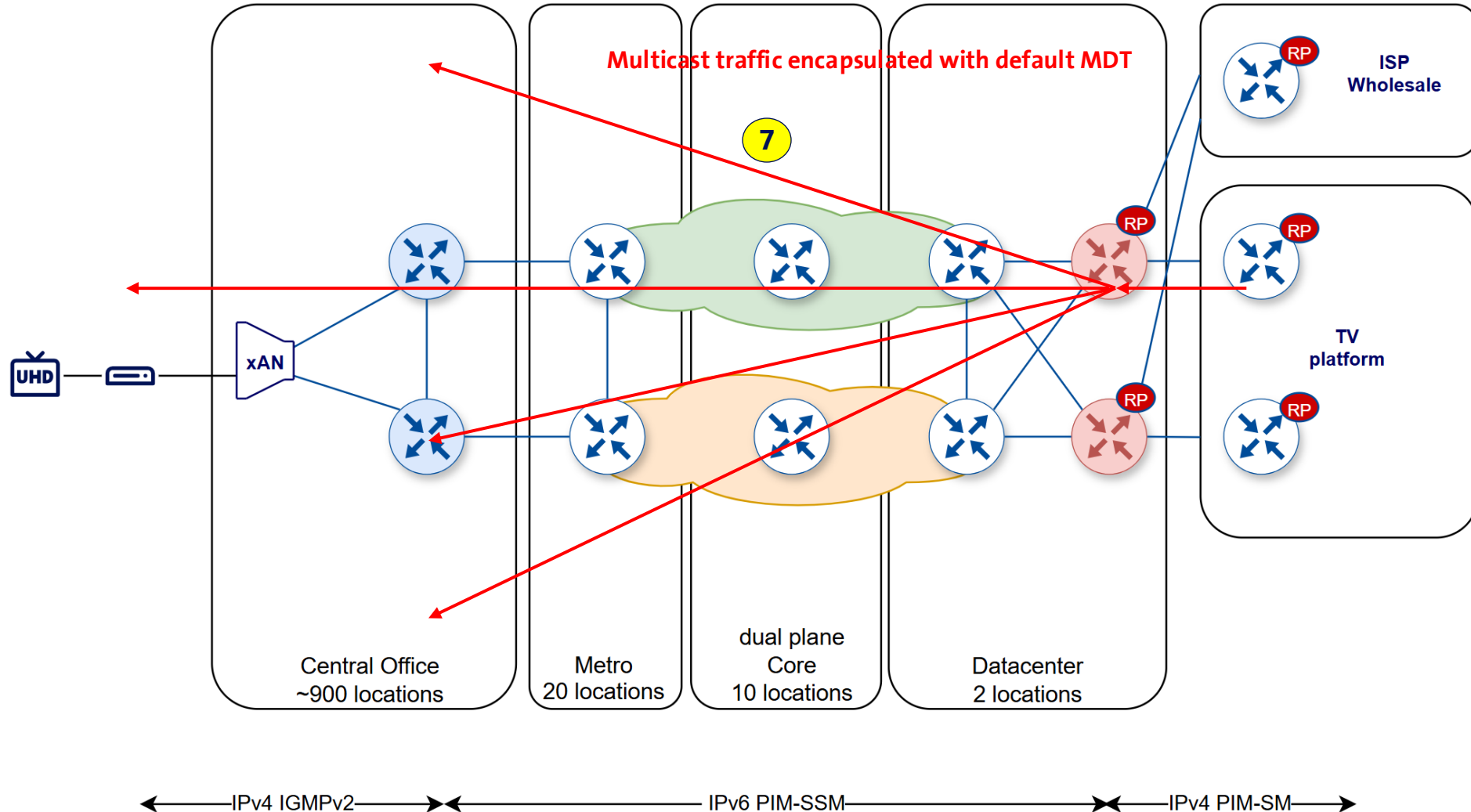
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Multicast flow setup - 7

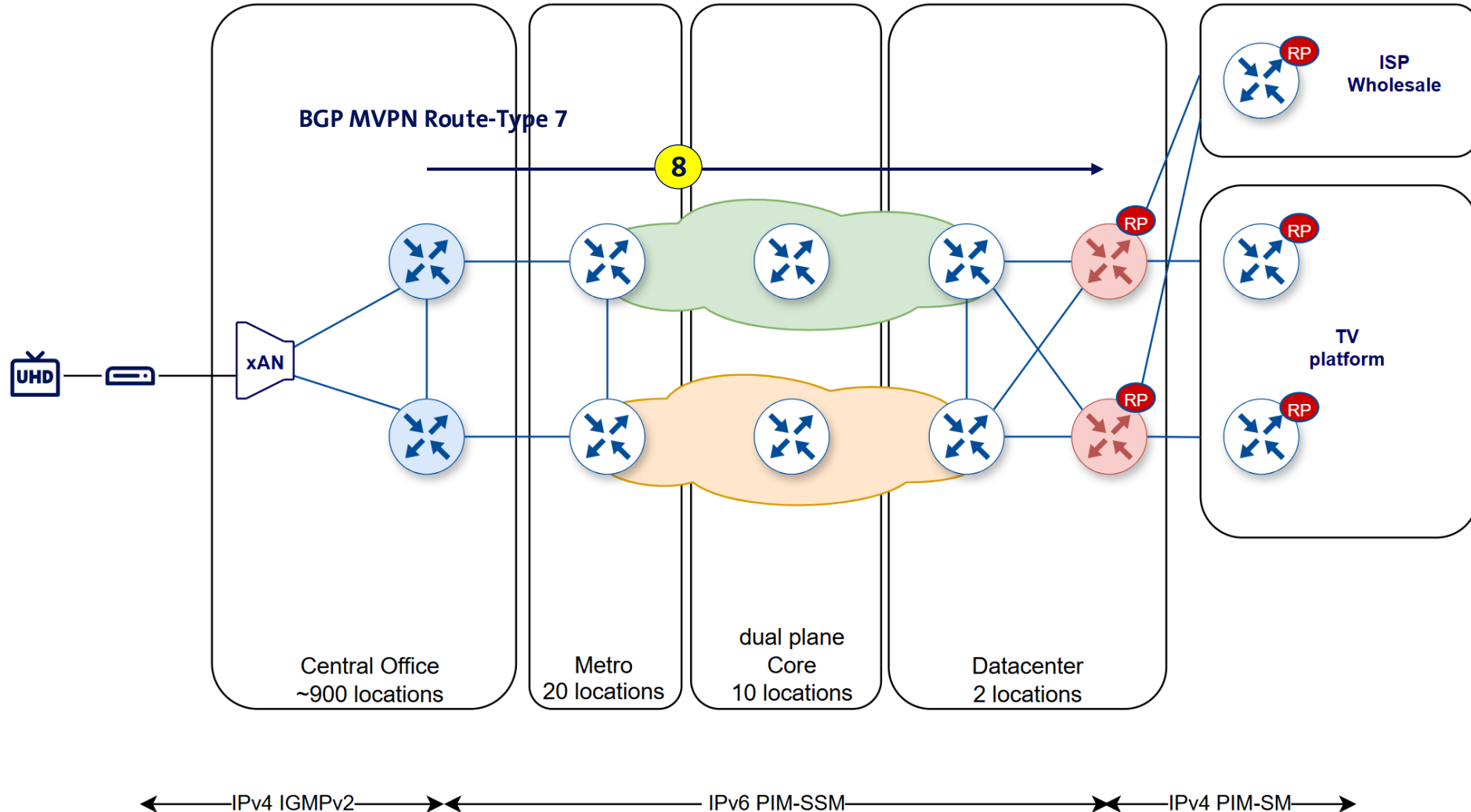
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Multicast flow setup - 8

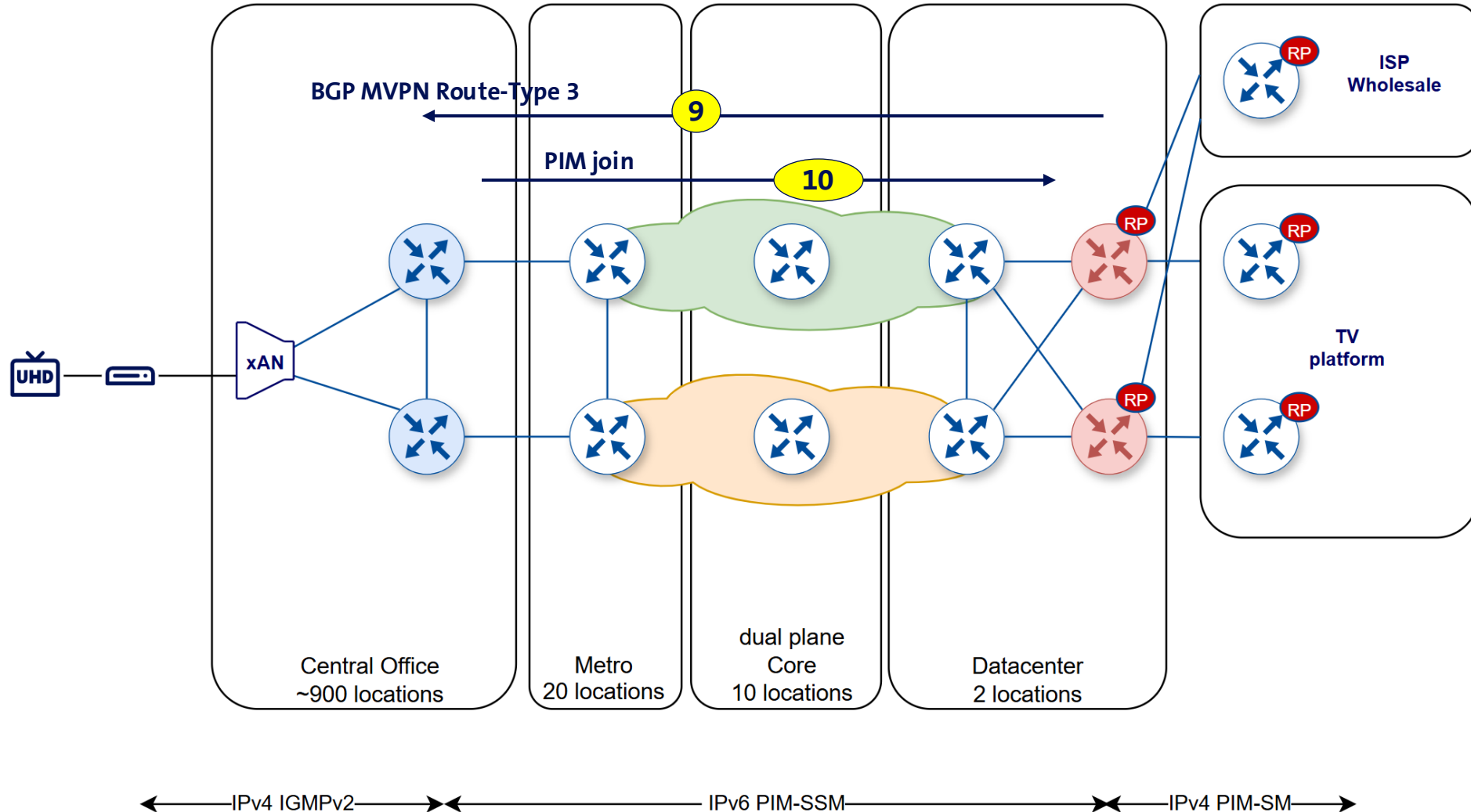
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Multicast flow setup – 9-10

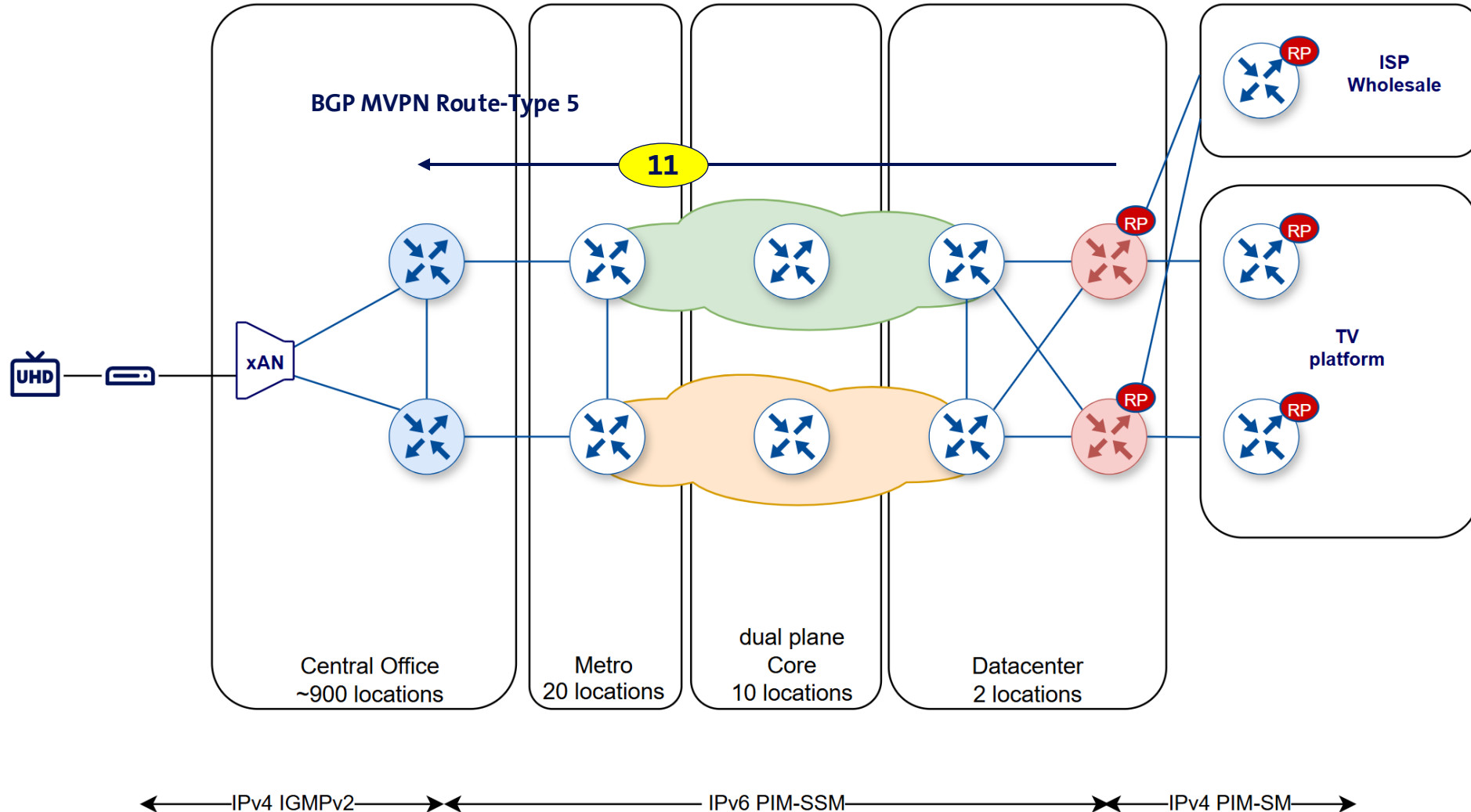
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Multicast flow setup - 11

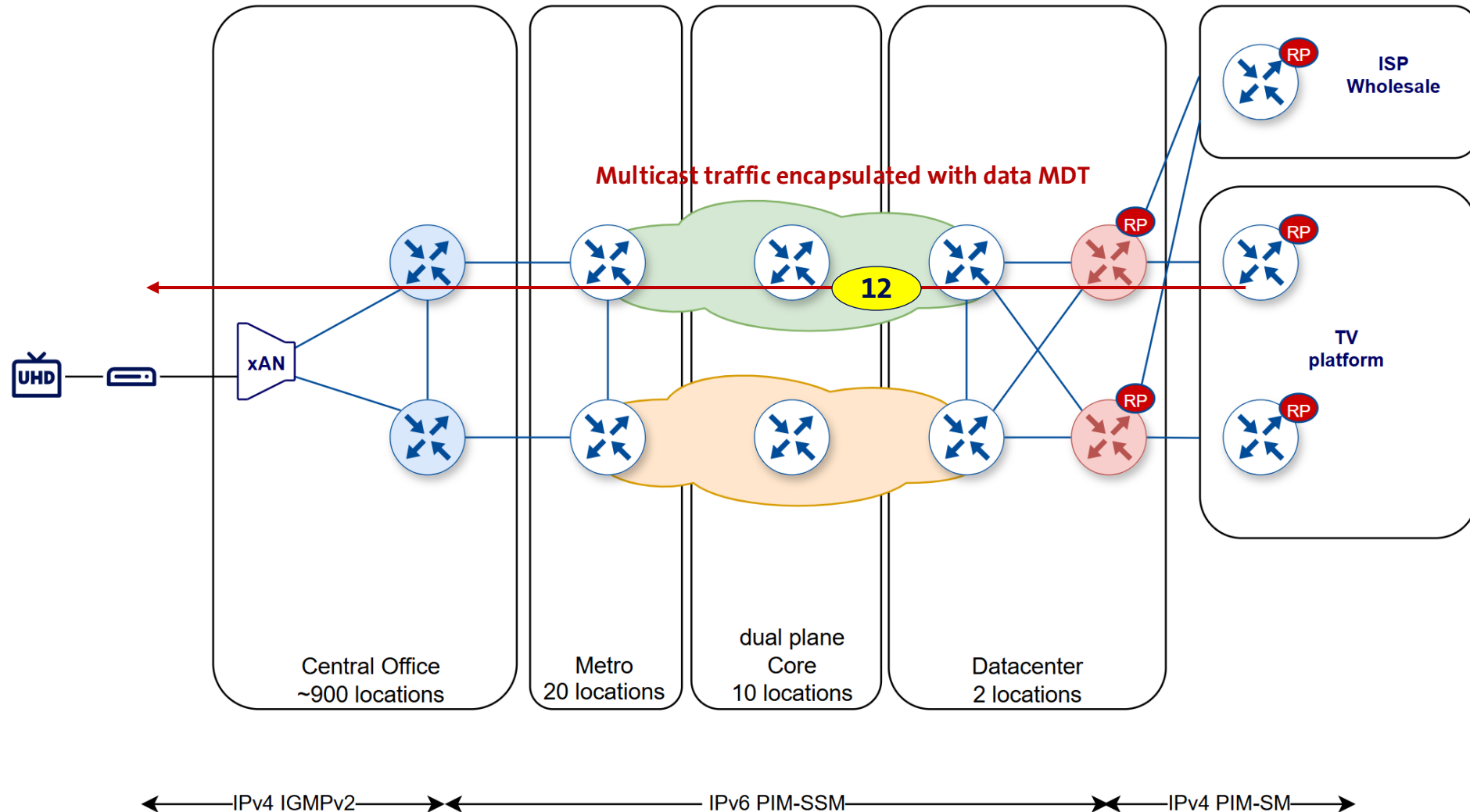
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Multicast flow setup - 12

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Ongoing work

- Fulfillment
 - Netconf validation (NEDs)
- Monitoring
 - adapting to mvpn/IPv6 core
- Further optimization:
 - MoFRR
 - EVPN DF election, load-balancing

Questions?

C1 Public



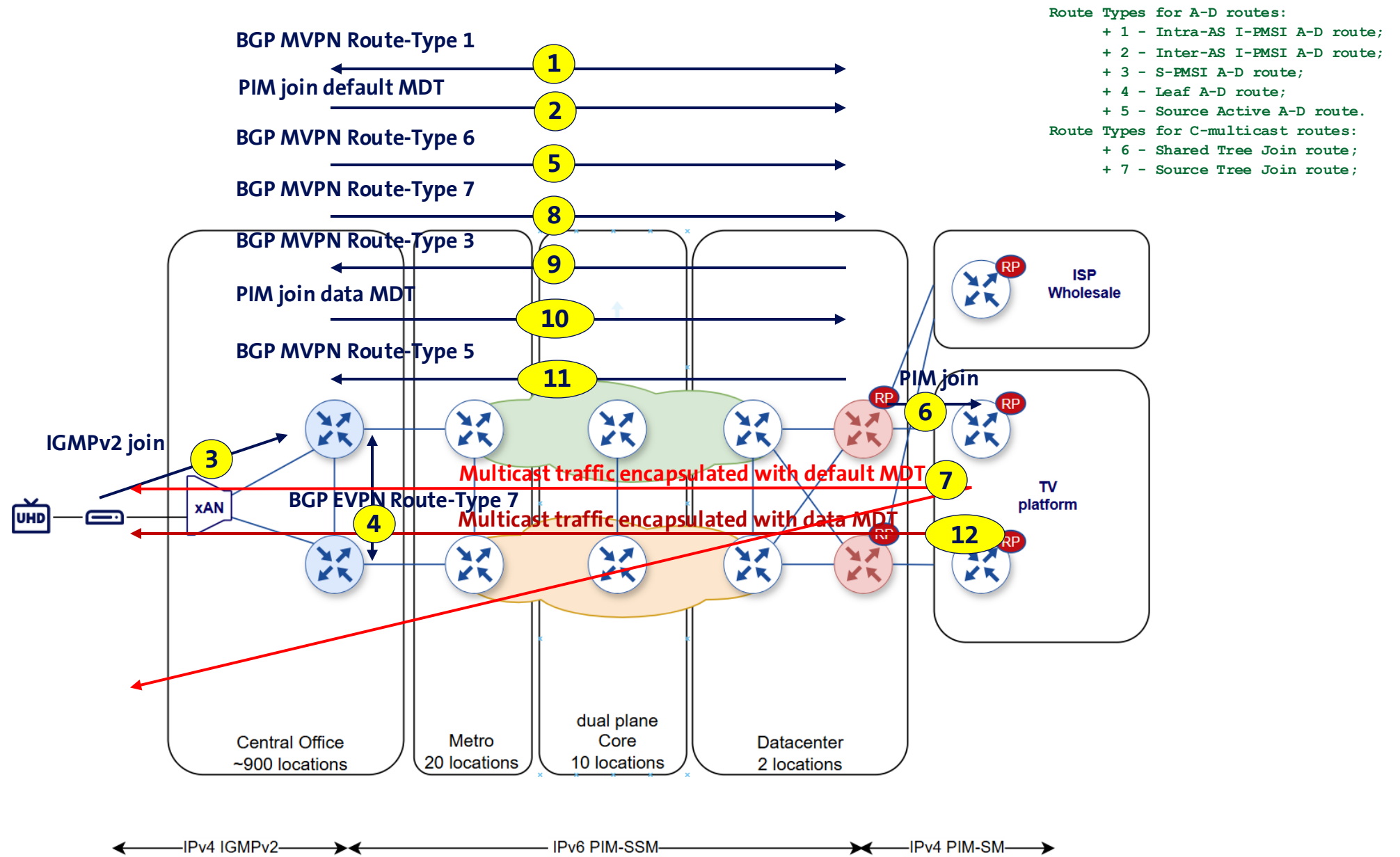
Thank you for your attention!





References

Title	Link
Multicast in MPLS/BGP IP VPNs	https://datatracker.ietf.org/doc/rfc6513/
BGP Encodings and Procedures for Multicast in MPLS/BGP IP VPNs	https://datatracker.ietf.org/doc/rfc6514/
Multicast over SRv6 networks	https://www.ietf.org/archive/id/draft-mankamana-pim-multicast-over-srv6-01.html
BGP Overlay Services Based on Segment Routing over IPv6 (SRv6)	https://datatracker.ietf.org/doc/rfc9252/
Internet Group Management Protocol (IGMP) and Multicast Listener Discovery (MLD) Proxies for Ethernet VPN (EVPN)	https://datatracker.ietf.org/doc/rfc9251/
Cisco Systems' Solution for Multicast in BGP/MPLS IP VPNs (historical Rosen draft)	https://datatracker.ietf.org/doc/rfc6037 https://www.cisco.com/c/en/us/support/docs/ip/multicast/200512-Configure-mVPN-Profiles-within-Cisco-IOS.html#anc29





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Classification	<...>
Scope of application	<...>
Document owner	<...>
Version	0.1
Issue date	<...>
Status	<...>
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Valid until	<...>
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Contact information

Swisscom Schweiz AG
Network connectivity
Network Architecture & Engineering
Mihai Ciurea
Alte Tiefenaustrasse 6
CH-3048 Worblaufen

+41 79 309 52 14
mihai.ciurea@swisscom.com